Ref. [34] — Convergence of Adam Under Relaxed Assumptions

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1. The problem and why the paper exists

Adam works remarkably well in large-scale deep learning, but a convergence proof is not automatic: adaptive denominators couple all previous stochastic gradients, and earlier proofs often assume globally bounded gradients or a uniform global smoothness constant. The authors ask whether one can prove convergence of Adam to *first-order stationary points* under a more realistic class of objectives, without assuming in advance that every gradient is bounded.

2. Start from the optimization objective and Adam update

The paper studies non-convex stochastic optimization

$$\min_{x\in\mathbb{R}^d} f(x), \qquad f(x)=\mathbb{E}_\xi[f(x;\xi)], \qquad \mathbb{E}[g_t \mid x_t]=\nabla f(x_t).$$

For orientation, the usual Adam statistics and update have the form

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2)(g_t \odot g_t), \\ x_{t+1} &= x_t - \eta_t \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t}+\delta}, \end{aligned}$$

with coordinatewise division and bias corrections. The denominator makes the effective step size coordinate dependent. These generic equations introduce the method; the paper's theorem requires its own parameter choices and technical conditions.

3. Why global smoothness and bounded gradients are restrictive

A classical assumption is $\|\nabla^2 f(x)\| \le L$ everywhere. The paper instead allows local curvature to grow with the gradient norm: its generalized smoothness class bounds the Hessian norm (where it exists) by a *sub-quadratic* function of $\|\nabla f(x)\|$. A schematic example of this idea is

$$\|\nabla^2 f(x)\| \le L_0 + L_1 \|\nabla f(x)\|^q, \qquad 0 \le q < 2,$$

where the exact regularity specification and constants are governed by the paper. This is *not* the assertion that every neural network satisfies the inequality globally.

Teaching illustration, not an experiment from the paper. For $f(x)=e^x$, $f'(x)=f''(x)=e^x$; no global constant L bounds the second derivative over $\mathbb{R}$, although curvature scales directly with gradient size. A proof can exploit such local information instead of requiring a single worst-case curvature bound.

4. Proof idea: bound the actual trajectory rather than assume it

The central technical innovation is an argument showing that *along the Adam iterates*, gradient magnitudes stay bounded with high probability under the specified generalized smoothness and oracle assumptions. Conceptually the proof first controls possible excursions into high-gradient/high-curvature regions, then establishes sufficient descent in expectation through the adaptive update. This is different from assuming $\sup_x \|\nabla f(x)\| < \infty$ for the entire domain.

> **Reported result (authors).** Adam finds an ϵ-stationary point, $\|\nabla f(x)\| \le \epsilon$, with $\mathcal{O}(\epsilon^{-4})$ stochastic-gradient complexity under the paper's relaxed conditions and parameter choices. The authors additionally analyze a variance-reduced Adam variant with $\mathcal{O}(\epsilon^{-3})$ complexity. These are *first-order* guarantees; they do not assert negative-curvature escape or global optimality.

5. Mathematical intuition behind the rates

A typical non-convex descent analysis tries to bound a sum of gradient norms:

$$\sum_{t<T} \eta_t \mathbb{E}\|\nabla f(x_t)\|^2 \;\lesssim\; f(x_0) - \inf f + \text{noise/adaptivity terms}.$$

For stochastic gradients, controlling the noise term can lead to T of order ϵ^{-4} for an ϵ-gradient target. Variance reduction makes gradient tracking more accurate as optimization proceeds, improving the oracle dependence to ϵ^{-3} in the paper's method. This display is an explanatory schematic, not the authors' verbatim theorem.

6. Assumptions, limits, and our survey's RQs

The relaxed curvature condition and stochastic-oracle controls remain substantive assumptions; the result does not justify unrestricted Adam hyperparameters on arbitrary losses. The bounded-trajectory conclusion is probabilistic and is not a global bounded-gradient property. There is no strict-saddle or SOSP statement. This work supports **RQ2** (adaptive optimization dynamics) and **RQ4** (relaxing unrealistic global smoothness) more directly than RQ3.

> **Do not claim:** "Adam is proved to escape all saddle points" or "Adam converges to global minima." The target is first-order stationarity, and finite-time complexity depends on the paper's assumptions.

7. Our notes

The contribution is not a new Adam update but a more realistic explanation of why Adam can converge. Instead of assuming gradients are bounded everywhere, the authors show how to control them along the actual trajectory, then obtain first-order rates. This is useful when discussing the gap between classical optimization assumptions and modern adaptive training. **Original source:** https://papers.neurips.cc/paper_files/paper/2023/hash/a3cc50126338b175e56bb3cad134db0b-Abstract-Conference.html.